

Evolving CAM16-UCS for Modular Accessible Perceptibility and Simultaneous Contrast Detection Tool for Geovisualization

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ABSTRACT

Accessibility is a foundational principle of effective visualization design, yet achieving perceptual accessibility remains a challenge for both experts and non-experts. Standards like the WCAG2 contrast ratio are intended to guide designers toward accessible color use but are increasingly misapplied as oversimplified and misunderstood shortcuts for achieving perceptual accessibility. This misuse highlights the need for more nuanced, perceptually grounded evaluation tools. In response, we present our Modular Accessible Perceptibility (MAP) contrast model and tool, which integrates advanced color appearance models to assess perceptibility and flag risks associated with simultaneous contrast. By augmenting the CAM16-UCS model with a CAM16 CCz simultaneous chromatic contrast calculation, our model can isolate and measure the predicted shift resulting from simultaneous contrast along the chroma axis. This enables a more refined evaluation of visual contrast that better reflects the complexities of human color perception. Additionally, our tool performs modularly segmented automated checks to assign MAP contrast index scores, offering a holistic assessment of a map's visual accessibility. This paper details the development and validation of the MAP model and tool, underscoring accessibility and inclusive design as fundamental rights, and demonstrating their utility in advancing perceptual accessibility in geovisualization.

Index terms: accessibility, perception, cartography, design, WCAG, CIE, color appearance model.

1 INTRODUCTION

The W3C WCAG2 web accessibility contrast ratio standard has long served as a way for cartographers, geovisualization developers and designers to systematically conform and improve design and visualization for an accessible online audience [1]. It is widely used by independent designers [2] and large industry players such as Apple alike [3]. Furthermore, WCAG standards serve as the basis for other important visualization tools and guidelines such as Chartability [4]. However, map, data and web designers often misunderstand the purpose and limitations of the standard, believing it as being all-purpose. In actuality, the WCAG2 contrast checker was meant for only one purpose: to evaluate the lightness difference between the colored text of a certain font size and style against the colored background [1]. Nonetheless, an overreliance on the WCAG2 contrast ratio standard seemingly took hold as the shortcut to achieve a supposed high “AA” or “AAA” accessibility rating without thorough understanding of what those rating means. This overreliance highlights the incomplete understanding of perceptual accessibility and the lack of better-tailored tools and resources for achieving higher visual accessibility in design.

However, this overreliance also highlights the fundamental importance of inclusive design and accessibility. Further, it underscores the need and well-intentioned pursuit for a high accessibility rating in geovisualization design.

Simply put, inclusive design and accessibility is a fundamental universal right and a pursuit that benefits all users and is worthy of attention and understanding. Accordingly, the W3C is in the process of developing and drafting the next version of its web accessibility standard (WCAG3) [5]. In WCAG3, the basic contrast ratio check is proposed to be superseded with a more comprehensive solution, one such proposal is the Accessible Perceptual Contrast Algorithm (APCA) score, though it has received considerable pushback [6]. The APCA uses a more progressive lightness contrast (Lc) compliance reference table that considers typeface geometry, color value (lightness) and other contextual factors [7]. This shift represents a much-needed evolution in how design accessibility is evaluated.

In that vein, this project aims to explore the utilization of advanced color appearance models (CAMs) to model and comprehensively measure perceptual color contrast along multiple perceptual color dimensions. The objective is to develop a more advanced domain-specific tool that facilitates the efficient evaluation of visual and perceptual contrast in geovisualization symbology. Ultimately, the project aims to create an open-access resource for cartographers and geo-visualization professionals that contributes to the updated WCAG standard and further extend open resources.

2 RELATED WORK

This project builds upon related work spanning accessibility, cognitive cartography, color psychology and perceptual color modeling.

2.1 Accessible Perceptibility Contrast

Perceptibility can be defined as the ability of an individual to clearly and easily perceive and distinguish different elements within a visual scene [8]. This encompasses the ability to distinguish between foreground and background elements, where factors such as simultaneous contrast can influence visual discrimination. In cartography, perceptibility is crucial for effective map reading and interpretation. Map users need to be able to distinguish between different geographical features represented by varying colors, patterns, and line weights. For example, different map stylesheet treatments might be represented by distinct colors, and the contrast between these colors and the background map needs to be sufficient for a user to easily differentiate them.

The W3C has developed standards and guidelines aimed at fostering a more accessible web environment, which inherently supports enhanced perceptibility of digital content [1]. These guidelines address aspects of visual design that impact the ease with which users can perceive and interact with web elements.

The WCAG2 contrast ratio widely employed to assess digital content accessibility, is dependent on only one dimension of color

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perception – lightness [9]. Lightness being how light or dark an element appears. Moreover, this ratio only considers the colors of text and its background with two breakpoints that reference typeface size and weight. To meet the highest level of accessibility standards, “AAA,” the contrast ratio of text to background needs to be 7:1 or above for text under 18 pt font size or 14 pt bolded type. If the text exceeds these setpoints, the contrast requirement for “AAA” rating is only 4.5:1. Though touted as a success in accessibility standards, this ratio fails to consider many considerations of human visual perception that influence perceptibility such as chroma, hue, and geometry (size, shape, weight, etc.), which is a concern for advanced data and geospatial visualization work.

The W3C is seeking to replace this contrast ratio in the forthcoming WCAG3 standards [5]. One proposed solution is the Acceptable Perceptual Contrast Algorithm (APCA) [7]. The APCA uses a more progressive (Lc) value that takes into consideration the luminance of the screen being used and factors in a more progressive consideration for the font’s size and weight [1]. The APCA represents a much-needed progression towards universal design principles but not without shortcomings and criticisms. Importantly, it does not take into consideration the color dimension of chroma, hue or the interaction of colors as a result of human visual perception dynamics, namely simultaneous contrast. Further improvements are therefore still needed to better account for the range of human perceptual ability and conditions.

2.2 Simultaneous Contrast Problem

Simultaneous contrast refers to the phenomenon where the perceived color of a stimulus changes due to the influence of surrounding attributes, in particular, color [10]. The current visual contrast accessibility standards in WCAG2 and WCAG3 insufficiently accounts for complex human vision dynamics. They fall short of addressing the ever-present simultaneous contrast problem. The induced shift in the perceived stimulus color occurs along the opponent color dimensions [11]. For instance, a light background can make a stimulus appear darker, while a dark background can make it appear lighter [12]. Similarly, as shown in Fig. 1 red induces green, green induces red, blue induces yellow and yellow induces blue. Neurologically, this phenomenon results from the structure of the opponent retinal photoreceptor cells [13]. Due to the role these retinal cells play in modulating stimuli light detection, this vision illusion results in both chromatic [14] and lightness value perceptual shifting [13]. This perceptual shifting consequently impacts overall perception of visual contrast [15].



Figure 1: Simultaneous contrast examples demonstrating the baseline color perception shift of identical inner colors due to their adjacent surrounding colors (left versus right).

Moreover, simultaneous contrast is impacted by the geometric dimensions and visual weight of a stimulus in context of the surrounding background [16]. Research by Grasso et al., has shown that with increasing visual complexity the impact of simultaneous contrast is intensified [17]. Due to the variable effects of simultaneous contrast on human color perception, it is challenging to develop an index that appropriately contextualizes its perceptual analogs. To account for simultaneous contrast, all four dimensions that influence color perception – chroma, hue, lightness, and

geometry must be modeled. The aforementioned WCAG2 and WCAG3 standards for predicting contrast do not substantively account for these dimensions. The inaccurate prediction of perceived color due to unaccounted simultaneous contrast can result in the measured contrast not aligning with the perceived contrast. This inaccuracy subsequently counters both the usability and accessibility of visual information. The risk associated with simultaneous contrast in geovisualization is that an otherwise well-designed color scheme may become ambiguous due to unforeseen color interactions resulting from the complexities of human visual perception [12]. This ambiguity in turn elevates the risk for misinterpretation of the geovisualization. Therefore, we were motivated to develop a tool and model that evaluates visual contrast based on a more thorough accounting of human color perception dynamics to ensure that accessible standards more accurately reflect actual visual accessibility needs.

2.3 CIE CAM16-UCS color appearance model

Several Color Appearance Models (CAMs) have been proposed and developed over the years that has resulted in incremental progress towards solving the simultaneous contrast problem. CAMs attempt to model human color perception, originally for evaluating display viewing conditions. These models either base their calculations on modeling the human vision system, accounting for cone dynamics and opponent theory or predication accuracy [11]. Significant progress has been made to better account for the influence of three dimensions of simultaneous contrast shifting (chroma, hue and lightness) [18]. However, while CAMs are effective at modeling human color perception, current models are unable to natively factor in the influence of a stimuli’s geometry (size, shape, weight, etc.). To enable measurement of perceptual distance, CAMs can be augmented and systematized into a Uniform Color Space (UCS) model. A UCS structures steps of chroma, hue, and lightness within a Euclidean space to represent equally perceptible and just-noticeable differences (JND) in color appearance to a human observer [19]. JND is a psychophysical concept outlining the threshold required for detectable perceptual change [20].

The CAM16-UCS is the most current color appearance model of human color perception [21]. CAM16 was introduced by the International Commission on Illumination (CIE) as an update to the CAM02 model which itself was an update to the CAM97s model. The model’s focus on perceptual uniformity helps to ensure that color differences are better consistently perceived by the human eye, regardless of the actual hue, providing a more reliable approximation of color uniformity compared to other models [22]. Although originally designed for display calibrations, CAMs are highly useful for designers who need a perceptually measurable color space that resembles human color discrimination. This distance, or the difference between color pairs, can be measured and represented using via Δ (delta) correlate values [19]. CAMs serve as an invaluable tool for inclusive digital visualizations and other applications where precise color representation and measurement are paramount.

Compared to CAM02, CAM16 addresses mathematical limitations and enhances the model’s efficiency and accuracy in representing human color perception [23]. Eq. 1 shows the calculation of Euclidian perceptual distance correlate ($\Delta E_{CAM16-UCS}$) between two colors within CAM16-UCS. For the purpose of our MAP model discussion, our augmented $\Delta E_{CAM16-UCS}$ is also simplified as ΔLab .

Eq. 1

$$\Delta E_{CAM16-UCS} = \sqrt{(L'_{bg} - L'_{fg})^2 + (a'_{bg} - a'_{fg})^2 + (b'_{bg} - b'_{fg})^2}$$

Where:

L' is perceived lightness
 a' is the degree to which a color appears red or green
 b' is the degree to which a color appears blue or yellow
 bg denotes the extracted background color
 fg denotes the extract foreground color

However, CAM16 only accounts for the achromatic aspects of simultaneous contrast - lightness and hue [11]. Accordingly, Zhu et al. proposed an adaptation to the CAM16 model aimed at predicting chromatic contrast effects built upon human studies investigating the influence of lightness, chroma, and hue [24]. Zhu et al.'s model references the Euclidean distance coordinates of CAM16 as a basis for their custom contrast $\Delta C'$ calculation, shown in Eq. 2. The output is then used in their $\Delta C'_{cc}$ equation to calculate the chromatic contrast, shown in Eq. 3 [25]. Our model's combined $\Delta C'_{cc}$ equation is shown in Eq. 4.

$$\Delta C' = \sqrt{(a'_2 - a'_1)^2 + (b'_2 - b'_1)^2} \quad \text{Eq. 2}$$

$$\Delta C'_{cc} = \frac{9.7536}{0.4396 + e^{-0.1232 \times \Delta C'_{bg-fg}}} - 4.343 \quad \text{Eq. 3}$$

$$\Delta C'_{cc} = \frac{9.7536}{0.4396 + e^{-0.1232 \times \sqrt{(a'_{bg} - a'_{fg})^2 + (b'_{bg} - b'_{fg})^2}}} - 4.343 \quad \text{Eq. 4}$$

3 MODULAR ACCESSIBLE PERCEPTIBILITY (MAP) CONTRAST MODEL AND TOOL

Our Modular Accessible Perceptibility (MAP) contrast model and tool is a further continuation of our exploratory proof-of-concept work to examine the feasibility of developing a custom-tailored open-sourced perceptibility model for evaluating map color perceptibility based on the CAM02-UCS [26].

That previous exploratory work builds upon the proposed APCA reference table logic but augments it further with perceptual contrast measurement referencing the CIE CAM02 Uniform Color Space (CAM02-UCS) model. In that previous implementation, the integration of CAM02-UCS enabled us to directly evaluate map color perceptibility derived not solely from luminance contrast ratios but rather a Δ value calculated from baseline perceptual distance along dimensions of human color perception. Further, we originally anticipated that the CAM02-UCS model would enable us to account for representative axes of colorblindness as well as the dimensions of simultaneous contrast [27]. Our proof-of-concept revealed limitations of the CAM02 model particularly in accurately and substantively capturing the dimensions of highly dynamic simultaneous contrast and its inescapable impact on overall perceptibility. That previous iteration also highlighted and reinforced the limits of trying to use text-centric WCAG2 contrast and APCA measures for non-textual purposes. Thus, our current iteration drops WCAG2 contrast and APCA entirely in favor of tailoring the tool specifically for non-textual color perceptibility evaluations of geovisualization design elements.

3.1 MAP Contrast Model and Index

Accordingly, we developed this Modular Accessible Perceptibility (MAP) contrast model using the more perceptually accurate CIE CAM16-UCS model. As discussed in the prior section, the CAM16 model represents a major progression in accurately modeling human color perception and specifically along the axes where simultaneous contrast occurs [11]. Further, by incorporating the CAM16 CCz ($\Delta C'_{cc}$) simultaneous chromatic contrast equation by Zhu et al., our augmented model can further isolate and measure

the model's predicted shift resulting from simultaneous contrast along the chroma axis [25]. Moreover, our MAP tool entirely drops WCAG3's proposed APCA due to its simplistic reliance on the non-dynamic static reference table and other limitations.

Our MAP model augments an existing CAM16-UCS model from the "Colour" Python library by NumFOCUS contributors and developers under open-source New BSD License terms [28]. We augment the model to expand and tailor its functions to better suit the domain-specific goal of this project. Our augmented model is capable of extracting RGB color space pixel values from any detected color pairing and converts it into CIE XYZ linear color space values then to CIE Lab for model calculation. This on-the-fly conversion is needed because it is not presently possible to natively remap RGB color space value directly onto the CAM16 model's Lab perceptual color space. The RGB to XYZ to Lab color conversion function is enabled by the same "Colour" Python library [28]. The model then calculates an overall Δ color perceptual distance value of the CAM16 Lab color pair within the CAM16-Uniform Color Space model (Eq. 1). This Δ value can then be used for our basic MAP rating calculation. Additionally, our model enhances this overall Δ Lab calculation by isolating the Chroma (C) distance value for simultaneous contrast modeling. We do this by converting the isolating the ΔC value from the Δ Lab output as part of the Zhu et al. $\Delta C'_{cc}$ calculation (Eq. 4). The $\Delta C'_{cc}$ output value can then be used to predict simultaneous contrast. Our high simultaneous contrast risk threshold of $<15 \Delta C'_{cc}$ is based on the observed threshold of the human subject validated chromatic simultaneous contrast shifting by Pei et al. [11].

Our augmented MAP model was incorporated into a Python tool that performs automated semi-randomized spot checks to grade maps' MAP Contrast index score. We modeled and defined our MAP Contrast index as a more progressive and comprehensive analogous but more comprehensive metric to the easily understandable existing WCAG's contrast rating. Our MAP contrast index ratings are:

- **High Perceptibility rating**—indicates a high overall perceptibility rating based on the computed overall MAP Contrast model Δ Lab score of ≥ 36 . This rating is analogous to WCAG's AAA rating.
- **Acceptable Perceptibility rating**—indicates a middling but acceptable MAP Contrast model Δ Lab score of ≥ 15 but < 36 . This rating is analogous to WCAG's AA rating.
- **Poor Perceptibility rating**—indicates a low overall MAP Contrast model Δ Lab score of < 15 denoting major perceptuality issues with the employed symbology.
- Additionally, based on the $\Delta C'_{cc}$ threshold, our index separately flags any detected **risk for simultaneous contrast** ($\Delta C'_{cc} < 15$).

3.2 MAP Contrast Tool

As shown in Fig. 2., our MAP tool works by letting users input a single or multiple map image (PNG) to compute a MAP contrast index report for each input map. The tool works by first running a neural network-based holistically-nested edge detection (HED) function to identify and exclude any map marginalia or interactive UI elements from the map frame [29]. This edge detection function enables our tool to be used for evaluating both traditional static maps as well as screenshots of interactive geovisualization. Using the isolated map frame, our tool then segments the map frame into nine equal-divided segments for evaluation. For each segment, it then runs a RGB pixel count function to identify and sample nine most numerous and therefore the largest RGB pixel clusters and the primary adjacent background to form contrast pairings. While our pixel count function is not a comprehensive technique to account for graphical geometry, it helps our model factor in the size of the

dominant colors and prioritize it for our index evaluation. Compared to the current process for WCAG2 contrast checks – often requiring designers manually extract, note, and then check RGB HEX codes using online resources – our tool automates and greatly streamlines the flow by simultaneously detecting multiple adjacent contrast pairs for evaluation. The tool then passes detected contrast pairs to the model for calculation. After calculating the individual Δ scores of each contrast pair, the tool averages the model scores of contrast pairs and rates it against our MAP contrast index rating of the evaluated map segment. Finally, the tool generates a HTML report of the model scores and index ratings. The report provides an overall summary of segment averages as well as detailed identification highlighting segments of concern and problematic color pairs yielding a poor index rating, as shown in Fig. 3.

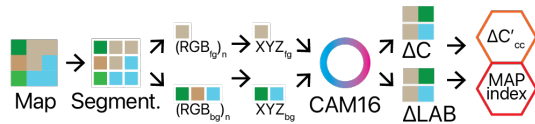


Figure 2: Simplified flow diagram of the MAP contrast tool.

4 DISCUSSION

Our MAP contrast model and tool are designed to assist cartographers and geovisualization professionals in evaluating perceptibility. Through this project, we were successful in this goal of prototyping a functional model and tool. Our prototype is able to provide users with an understandable and comparatively comprehensive perceptual evaluation of their map and geovisual product. By streamlining both the pipeline (tool) and evaluation (model), our prototype is markedly more comprehensive than the existing WCAG2 contrast standard and the proposed APCA algorithm. Nonetheless, our tool does not entirely seek to replace WCAG2 contrast standard. WCAG2 contrast and its to-be-determined WCAG3 successor are still generally sufficient for their intended text contrast evaluation purpose. What our MAP model and tool seek to solve is the inappropriate reliance on WCAG2 contrast for geovisual design evaluation as a shortcut for supposed high accessibility rating. Our tool does that by providing a tailored solution for this exact design use case. However, one area of ongoing work prior to public release is the need to validate our model against human subject ratings. We plan to publicly release the human validated version of our tool as an open-sourced solution hosted on our institutional GitLab site under a Creative Commons Attribution-Noncommercial (CC BY-NC) license.

The Δ values used for our index are based on the existing assumptions of modeling colors within Euclidian uniform color spaces. Our perceptibility index ratings were chosen based on that simplified computation of the color spaces. However, these values are not infallible analogs for actual human color perception. Rather, they represent our and others' interpretations of prior and current research. Our high perceptibility Δ Lab value of ≥ 36 assumes an approximate perceptual boundary in our custom model where contrast differences transition from being subtle to clearly noticeable for the average observer [19]. At the low end, our poor perceptibility Δ Lab value of < 15 roughly coincides with the perceptual limit for just-noticeable differences (JND) [11]. Our acceptable rating falls between these thresholds. These cited threshold, custom index rating, augmented model and tool are themselves guidelines derived from computation models and are not definitive rules. The perception of color difference and contrast is complex and is influenced by a myriad of factors beyond the limits of existing computational color models. Therefore, validation work is crucial for continued development of this project. To that effect,

we plan on conducting a follow-up validation study to further examine the suitability of our Δ value-derived index thresholds with actual visual judgement made by human observers.

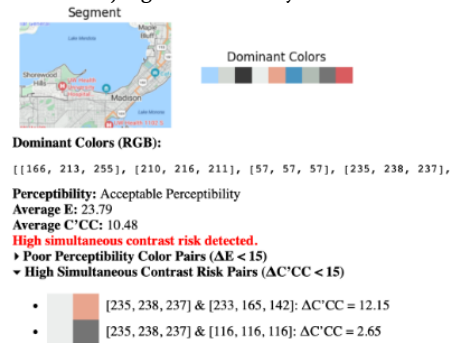


Figure 3: A portion of the MAP report with an overview of the evaluated segment. This report shows an overall acceptable average perceptibility rating but a high risk for simultaneous contrast from employed color pairs.

Another crucial aspect for our planned human subject validation study is the recognition and ongoing evolution in foundational cognitive psychology, which has moved away from the assumption of a neurotypical universal baseline for human perception and cognition. Contemporary scholarship is gradually shifting towards a more neurodiverse understanding and acceptance that human perception is subjective, even for fundamental functions like color visual perception. This shift underscores the potential pitfall and limitation of models and tools like ours. It is no longer acceptable to assume that this model can accurately capture universal baseline color perception. Instead, our model, along with others, aims to replicate the neurotypical experience as a valuable empirical standard but for specific purposes.

A functional limitation of our MAP model is the limited incorporation of geometric factors including shape and weight. As discussed, while our pixel cluster function identifies the largest graphical clusters to prioritize evaluation, our model calculation presently does not factor in the perceptual impact stemming from geometric differences. This limit stems from the project's limited scope, which focused on advancing perceptual contrast calculation and simultaneous contrast prediction using known perceptual baseline metrics referenced to the CAM16-UCS color appearance model. Therefore, to account for geometric influences, additional research is necessary to measure baseline shifts within our color appearance model caused by geometric differences. However, this research was beyond the scope of our original project. Nevertheless, we are considering its feasibility as part of our planned follow-up human subject study.

5 CONCLUSION

This project introduces a framework, processual model, and tool designed to advance how cartographers, web developers, and visualization experts approach and evaluate design accessibility. While not intended as a comprehensive solution to universal perceptual accessibility, it represents a meaningful step toward expanding the pathways available for inclusive design—specifically through the lens of color contrast and visual perceptibility.

By bridging the gap between perceptual theory and practical application, our work equips the geovisualization community with the tool to create more accessible, thoughtful, and inclusive geovisual products in line with open inclusive design standards.

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REFERENCES

- [1] "Web Content Accessibility Guidelines (WCAG) 2.0." Accessed: Apr. 23, 2025. [Online]. Available: <https://www.w3.org/TR/WCAG20/#visual-audio-contrast>
- [2] A. Tickner, "Accessibility in map design," in *Annual Meeting of the North American Cartographic Information Society*, Minneapolis: NACIS, Oct. 2022. Accessed: Apr. 24, 2024.
- [3] Apple Inc., "Accessibility | Apple Developer Documentation." Accessed: Apr. 24, 2024. [Online]. Available: <https://developer.apple.com/design/human-interface-guidelines/accessibility#Color-and-effects>
- [4] F. Elavsky, C. Bennett, and D. Moritz, "How accessible is my visualization? Evaluating visualization accessibility with Chartability," *Computer Graphics Forum*, vol. 41, no. 3, pp. 57–70, Jun. 2022, doi: 10.1111/CGF.14522.
- [5] World Wide Web Consortium, "WCAG 3 Introduction | Web Accessibility Initiative (WAI) | W3C." Accessed: Apr. 24, 2024. [Online]. Available: <https://www.w3.org/WAI/standards-guidelines/wcag/wcag3-intro/>
- [6] World Wide Web Consortium, "Visual contrast of text | How-To | WCAG 3 | Web Accessibility Initiative (WAI) | W3C." Accessed: Apr. 24, 2024. [Online]. Available: <https://www.w3.org/WAI/GL/WCAG3/2020/how-tos/visual-contrast-of-text/>
- [7] K. Ulitin, "Applying APCA and Huetone for Color Accessibility of User Interfaces," *Communications in Computer and Information Science*, vol. 1833 CCIS, pp. 382–388, 2023, doi: 10.1007/978-3-031-35992-7_53.
- [8] A. Kim, H. S. Kim, and S. O. Park, "Measuring of the perceptibility and acceptability in various color quality measures," *J Opt Soc Korea*, vol. 15, no. 3, pp. 310–317, Sep. 2011, doi: 10.3807/JOSK.2011.15.3.310.
- [9] "Web Content Accessibility Guidelines (WCAG) 2.2." Accessed: Apr. 30, 2025. [Online]. Available: <https://www.w3.org/TR/WCAG22/>
- [10] M. D. Fairchild, "Color Appearance Phenomena," in *Color Appearance Models*, Wiley, 2013, pp. 115–141. doi: 10.1002/9781118653128.ch6.
- [11] E. Pei, S. Farnand, and M. D. Fairchild, "Extending CIECAM16 to Account for Simultaneous Contrast," *Color Res Appl*, 2025, doi: 10.1002/col.22980.
- [12] C. A. Brewer, "Prediction of simultaneous contrast between map colors with Hunt's model of color appearance," *Color Res Appl*, vol. 21, no. 3, pp. 221–235, 1996, doi: 10.1002/(SICI)1520-6378(199606)21:3<221::AID-COL4>3.0.CO;2-U.
- [13] E. H. Adelson, "Lightness Perception and Lightness Illusions," in *The New Cognitive Neurosciences*, 2nd ed., M. Gazzaniga, Ed., Cambridge: MIT Press, 2000, ch. 24, pp. 339–351. Accessed: Jul. 01, 2025.
- [14] K.-F. Yang and Y.-J. Li, "Double opponency serves as a basis for color constancy," Oct. 2024, Accessed: Apr. 29, 2025. [Online]. Available: <http://arxiv.org/abs/2410.08823>
- [15] L. E. Wool, J. D. Crook, J. B. Troy, O. S. Packer, Q. Zaidi, and D. M. Dacey, "Nonselective wiring accounts for red-green opponency in midget ganglion cells of the primate retina," *Journal of Neuroscience*, vol. 38, no. 6, pp. 1520–1540, Feb. 2018, doi: 10.1523/JNEUROSCI.1688-17.2017.
- [16] V. Shi, J. Cui, X. G. Troncoso, S. L. Macknik, and S. Martinez-Conde, "Effect of stimulus width on simultaneous contrast," *PeerJ*, vol. 2013, no. 1, 2013, doi: 10.7717/peerj.146.
- [17] P. A. Grasso, F. Tommasi, R. Franconi, E. Baldanzi, A. Farini, and M. Gurioli, "Simultaneous Color Contrast Increments with Complexity and Identity of the Target Stimulus," *Life*, vol. 15, no. 2, Feb. 2025, doi: 10.3390/life15020257.
- [18] Y. Zhu, D. Tian, and M. R. Luo, "A new independent dataset to verify the performance of colour appearance models for predicting simultaneous contrast effect," *Color and Imaging Conference*, vol. 28, no. 1, pp. 361–365, 2020, doi: 10.2352/issn.2169-2629.2020.28.58.
- [19] W. S. Mokrzycki and M. Tatol, "Colour difference ΔE - a survey," *Machine Graphics & Vision International Journal*, vol. 20, no. 4, pp. 383–411, Apr. 2011, doi: 10.5555/3166160.3166161.
- [20] X. Zhang, W. Lin, and P. Xue, "Just-noticeable difference estimation with pixels in images," *J Vis Commun Image Represent*, vol. 19, no. 1, pp. 30–41, Jan. 2008, doi: 10.1016/J.JVCIR.2007.06.001.
- [21] M. D. Fairchild, "A revision of CIECAM97s for practical applications," *Color Res Appl*, vol. 26, no. 6, pp. 418–427, Dec. 2001, doi: 10.1002/COL.1061.
- [22] R. M. Lou, G. Cui, and C. Li, "Uniform colour spaces based on CIECAM02 colour appearance model," *Color Res Appl*, vol. 31, no. 4, pp. 320–330, Aug. 2006, doi: 10.1002/COL.20227.
- [23] M. R. Luo *et al.*, "A comprehensive test of colour-difference formulae and uniform colour spaces using available visual datasets," *Color Res Appl*, vol. 48, no. 3, pp. 267–282, May 2023, doi: 10.1002/COL.22844.
- [24] Y. Zhu and M. R. Luo, "Proposed modification to the CAM16 colour appearance model to predict the simultaneous colour contrast effect," *Color and Imaging Conference*, vol. 27, pp. 190–194, Oct. 2019, doi: 10.2352/ISSN.2169-2629.2019.27.34.
- [25] Y. Zhu and M. R. Luo, "Modelling and modification of simultaneous lightness contrast effect using the Albers' pattern," *London Imaging Meeting*, vol. 1, no. 1, pp. 28–32, Sep. 2020, doi: 10.2352/ISSN.2694-118X.2020.LIM-29.
- [26] P. W. Limpisathian, "Evolving the WCAG3 Accessible Contrast Perceptibility Standard for Maps," in *Abstracts of the International Cartographic Association*, Vancouver, Aug. 2025.
- [27] M. R. Luo and C. Li, "CIECAM02 and Its Recent Developments," in *Advanced Color Image Processing and Analysis*, C. Fernandez-Maloigne, Ed., New York: Springer Science+Business, 2013, ch. 2, pp. 19–58. doi: 10.1007/978-1-4419-6190-7.
- [28] T. Mansencal *et al.*, "Colour," Oct. 11, 2024, *NumFOCUS* 0.4.6. doi: 10.5281/ZENODO.13917514.
- [29] S. Xie and Z. Tu, "Holistically-Nested Edge Detection," *Int J Comput Vis*, vol. 125, no. 1–3, pp. 3–18, Apr. 2015, doi: 10.1007/s11263-017-1004-z.